

DRIFT-SENSE

Resilient Localization. Uncompromised Precision.

AI-powered navigation-error recovery for periodic semiconductor wafer inspection.

APPLIED MATERIALS

FINFET-STYLE

DEEP LEARNING

TEAM DETAILS:

TEAM NAME: MCTE Team 2

COLLEGE NAME: MILITARY COLLEGE OF TELECOMMUNICATION ENGINEERING
MHOW, INDORE(M.P).

S/N	NAME	ROLE	CONTACT	
			MOBILE	E-MAIL
1.	MANAS BAJPAI	TEAM CAPTAIN, AI/ML Engineer		vlsimcte@gmail.com
2.	SUKHVEER SINGH	DATASET ENGINEER	8755700791	rahulkumarnauhwar179@gmail.com
3.	BOREY DUTTA	TESTING & DEPLOYMENT	8699615162	dattaborey2@gmail.com
4.	SUDHEER CHOUDHARY	DATASET DEVELOPMENT	8319795223	Jat.sudhir6677@gmail.com

THE INDUSTRY PROBLEM

A wafer inspection tool must revisit the same site repeatedly with extremely high positional consistency.

Small stage errors can accumulate through thermal expansion, vibration and mechanical slack. A revisit may land several pixels away from the intended site.

The difficulty is not only drift: semiconductor layouts repeat. A wrong location can look almost identical to the correct one.

WHY TEMPLATE MATCHING BREAKS

Highly repetitive DRAM cell arrays create multiple visually similar locations. A conventional template matcher may therefore lock onto the wrong periodic copy instead of the intended site.

OUR OBJECTIVE

Given a reference image and a larger search image:

1. Find the intended pattern location
2. Suppress periodic false matches
3. Remain robust to independent sensor noise, blur, rotation and scale variation
4. Output one center coordinate: (x, y)

1 • ARCHITECTURE

DRAM-STYLE

Synthetic patterns model **repetitive DRAM memory-cell arrays**, including:

- Word-line-like structures
- Bit-line-like structures
- Repeated memory-cell patterns
- Contact/storage-node features

Why DRAM?

- Strong periodic structure
- High visual similarity between neighboring cells
- Naturally exposes periodic localization ambiguity
- Suitable for testing robust navigation-error recovery

2 • LOCALIZATION

AI-BASED LOCALIZATION

A shared feature encoder extracts structural representations from both images.

Multi-scale correlation identifies candidate positions while preserving spatial context.

A **coordinate-refinement stage** estimates the final (x, y) center with sub-pixel precision.

3 • WHY NOT TEMPLATE MATCHING?

Raw template matching is dominated by repeated local appearance.

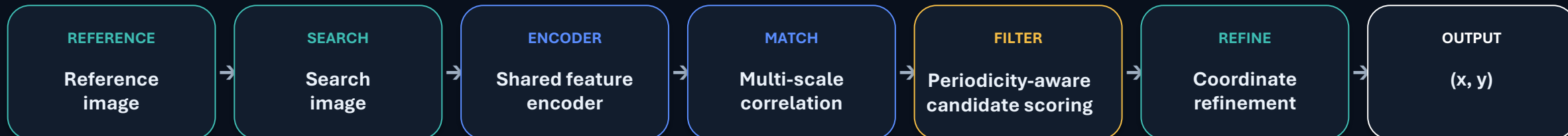
Drift-Sense instead combines:

- Learned structural features
- Multi-scale agreement
- Candidate consistency
- Coordinate refinement

Goal: Identify the intended DRAM cell location—not simply the most visually similar location.

Proposed Solution

Controlled synthetic data + periodicity-aware multi-scale localization



DATASET GENERATOR

- DRAM-style periodic cell structures
- Wordline/bitline-like repeating patterns
- Randomized cell placement and pitch
- Reference & search image size variation
- Independent sensor noise
- SEM-inspired edge brightening/contrast
- Blur, rotation & scaling

Controlled variation allows the model to learn **realistic localization uncertainty** rather than memorize fixed patterns.

LOCALIZATION MODEL

- Shared-weight feature encoder
- Multi-scale feature pyramid
- Reference-to-search correlation
- Candidate peak extraction
- Periodicity-aware suppression
- Final coordinate refinement

Design principle: Use structural and contextual evidence across scales instead of relying only on raw pixel similarity.

ROBUSTNESS DESIGN

Training emphasizes variation rather than memorization.

Test protocol:

- ≥ 30 fresh randomized pairs
- Higher search-image noise than training
- Both easy and highly periodic regions
- Report error distribution, tolerance accuracy and runtime

This mirrors the official Applied Materials test intent [1].

Innovation & Uniqueness

Designed around the actual failure mode: periodic ambiguity

01 Periodicity-Aware Scoring

Avoids the assumption that the highest local similarity is unique. Candidate locations are compared using structural and multi-scale evidence.

02 Independent-Capture Simulation

Reference and search images receive independent sensor noise, reflecting separate physical acquisitions.

03 SEM-Informed Image Formation

Edge-brightening is explicitly modeled because secondary-electron SEM images show characteristic bright edge contrast.

04 10× Scale-Aware Localization

The system is designed for the required reference/search scale relationship using multi-scale features and coordinate refinement.

Use measured results from the exact GitHub version submitted to i4C

TEST CASES

[30]

≤1 px ACCURACY

[100%]

≤3 px ACCURACY

[100%]

≤5 px ACCURACY

[100%]

MEAN ERROR

[0.0 px]

INFERENCE

[397.25 ms]

Evaluation performed on the current 30-sample DRAM v2 evaluation set.

SUCCESS CASE — INSERT ACTUAL VISUAL

Reference image | Search image

True center: (646,408)

Predicted center: (646,408)

Pixel error: [0.00 px]

Matching score : (0.9202)

Successful localization under rotation and scale variation.
The proposed rotation-scale search followed by full-resolution verification correctly localized the DRAM target with zero pixel error.

HONEST FAILURE — INSERT ACTUAL VISUAL

Reference image | Search image

True center: (154,813)

Predicted center: (173, 522)

Pixel error: [291.62 px]

Failure caused by periodic structural ambiguity.

The search image contained visually similar repeating DRAM patterns. The matcher selected a structurally similar distractor instead of the true target location. This failure motivated the final rotation-scale locked local verification approach.

SOFTWARE STACK

Python 3.14.2
OpenCV 5.0
NumPy 2.4.1
SciPy 1.17
Pandas 3.0.1
Matplotlib 3.9+
scikit-learn 1.8

Model export: PyTorch / ONNX
(optional)
CLI inference: standalone Python
script

DEVELOPMENT HARDWARE

GPU: [Actual GPU]
CPU: [Actual CPU]
RAM: [Actual RAM]
CUDA: [Version]
Cloud: [None / Platform]

Dataset generation: [~Few minutes /
30 samples]
Training: [time]
Model size: [N/A]
Inference: [397.25 ms / pair]

FEASIBILITY & SCALABILITY

- Generator parameters expose architecture, noise and placement diversity.
- Ground truth is generated automatically, enabling repeatable evaluation.
- Inference is packaged as a single command for evaluator use.
- The architecture can be extended from FinFET to DRAM-style layouts without changing the core coordinate-output interface.

PUBLIC GITHUB REPOSITORY

Repository:

<https://github.com/dattaborey2-ai/Drift-Sense-Wafer-Inspection>

Required contents:

- README.md
- generate_dataset.py
- inference.py ← critical evaluator script
- train.py / notebook
- model weights (.pt/.onnx) if DL
- requirements.txt
- ground_truth.csv / evaluation code
- references / citation document
- example input pair

EVALUATOR RUN

Dataset generation:

```
python generate_dataset.py --architecture finfet --num_pairs 100 --  
output_dir ./dataset
```

Inference:

```
python inference.py --reference reference.png --search search.png
```

Output:

(x, y)

No manual path edits, GUI steps or source-code changes.

Use the same numbering in the PPT, README and citation document

[1] i4C / SEMICON India Hackathon 2026 — Drift-Sense problem statement and submission requirements.

Official

[2] Villarrubia, J. S. et al. (2001), “Edge Determination for Polycrystalline Silicon Lines on Gate Oxide,” NIST / SPIE.

SEM edge

[3] ETH Zurich ScopeM, “SEM – Imaging with Secondary Electrons.”

SEM edge

[4] Microelectronic Engineering (2019), “Deep learning denoising of SEM images towards noise-reduced LER measurements.”

SEM noise

[5] Szeliski, R., Computer Vision: Algorithms and Applications, Springer.

CV

[6] Lowe, D. G. (2004), “Distinctive Image Features from Scale-Invariant Keypoints,” IJCV.

Scale/registr
ation

[7] Oxford Academic, “Spatial resolution in secondary-electron microscopy.”

SEM imaging